The Nernst Equation and Cell Potential

Answer key with worked solutions

A. Ideas

1. Battery voltage.

> As the cell runs, reactants are used up and products build up, so the reaction quotient **Q increases**. By the Nernst equation, $E = E^\circ - \frac{0.0592\ \text{V}}{n}\log Q$, so E **decreases**. When the reaction reaches equilibrium, Q = K and **E = 0**: the battery is flat.

B. Using the Nernst equation

2. Zn–Cu cell.

> $E^\circ = 1.10\ \text{V}, n = 2, Q = \dfrac{0.10}{2.0} = 0.050$
>
> $$E = 1.10\ \text{V} - \frac{0.0592\ \text{V}}{2}\log 0.050$$
> $$= 1.10\ \text{V} + 0.0385\ \text{V} = \mathbf{1.14\ V}$$
>
> **Larger** than E°: Q is less than 1 (more reactant Cu^{2+} than product Zn^{2+}), so the reaction has a greater driving force.

3. Ag–Cu cell.

> (a) $E^\circ = 0.80\ \text{V} - 0.34\ \text{V} = \mathbf{0.46\ V}$; $n = \mathbf{2}$ ($2\ Ag^+$ each gain $1\ e^-$)
>
> (b) $Q = \dfrac{[Cu^{2+}]}{[Ag^+]^2} = \dfrac{1.0}{(0.010)^2} = 1.0 \times 10^4$
>
> $$E = 0.46\ \text{V} - \frac{0.0592\ \text{V}}{2}\log(1.0 \times 10^4)$$
> $$= 0.46\ \text{V} - 0.118\ \text{V} = \mathbf{0.34\ V}$$

4. Silver concentration cell.

> (a) The **dilute** side (0.0010 M) is the anode: Ag is oxidized there, raising its Ag^+ concentration.
>
> (b) $E^\circ = 0, n = 1, Q = \dfrac{0.0010}{0.10} = 0.010$; $E = 0 - 0.0592\ \text{V} \times \log 0.010 = \mathbf{+0.118\ V}$

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C. E°, ΔG° and K

5. ΔG° and K.

> (a) $\Delta G^\circ = -2 \times 96\,485\ \text{C/mol} \times 0.46\ \text{V} = -8.88 \times 10^4\ \text{J/mol} = \mathbf{-88.8\ kJ/mol}$
>
> (b) $\log K = \dfrac{2 \times 0.46\ \text{V}}{0.0592\ \text{V}} = 15.54$, so $K = \mathbf{3.5 \times 10^{15}}$

6. E° from K.

> $E^\circ = \dfrac{0.0592\ \text{V}}{1} \log(1.0 \times 10^5) = 0.0592\ \text{V} \times 5.0 = \mathbf{0.30\ V}$

7. Negative E°.

> $\Delta G^\circ = -nFE^\circ$ is **positive**, and K is **less than 1**: the reaction is not spontaneous under standard conditions, and reactants are favoured at equilibrium.

D. Challenge

8. When E = 1.00 V.

> $$1.00\ \text{V} = 1.10\ \text{V} - \frac{0.0592\ \text{V}}{2} \log Q$$
> $$\log Q = \frac{2 \times 0.10\ \text{V}}{0.0592\ \text{V}} = 3.38$$
> $$Q = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = \mathbf{2.4 \times 10^3}$$
>
> The voltage falls only slowly at first: zinc ions must outnumber copper ions by about 2400 to 1 before E drops by 0.10 V.

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